

Encryption And Hashing

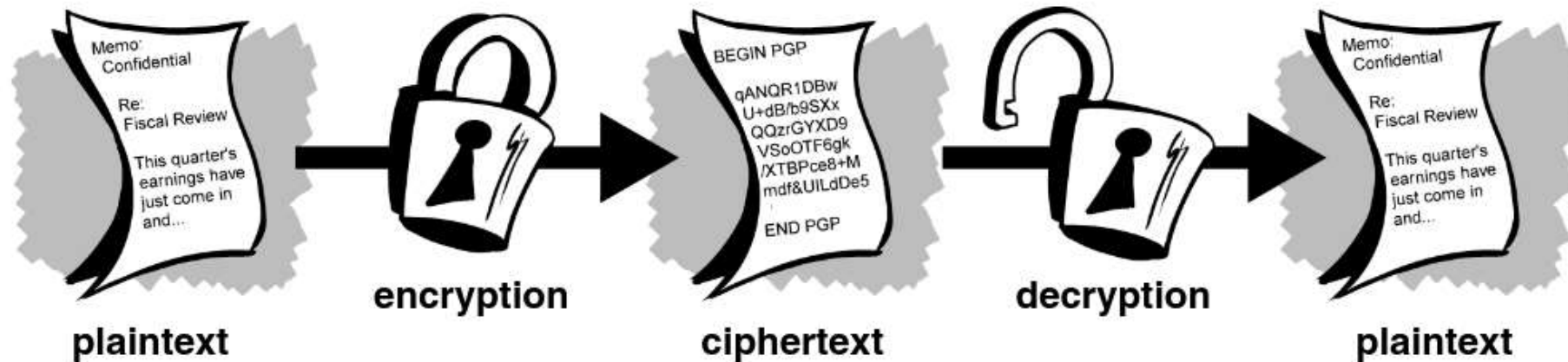
Cryptography

- Cryptography is the science related to encrypt and decrypt the data to make it more secure
- To encrypt you use a key and to decrypt you use a key, the key for encryption and decryption can be the same and can be different

Encryption



- Encryption is to tie the letter of secrets with a key then send it to the receiver and he can decrypt it with the key he owns



What is cryptography?

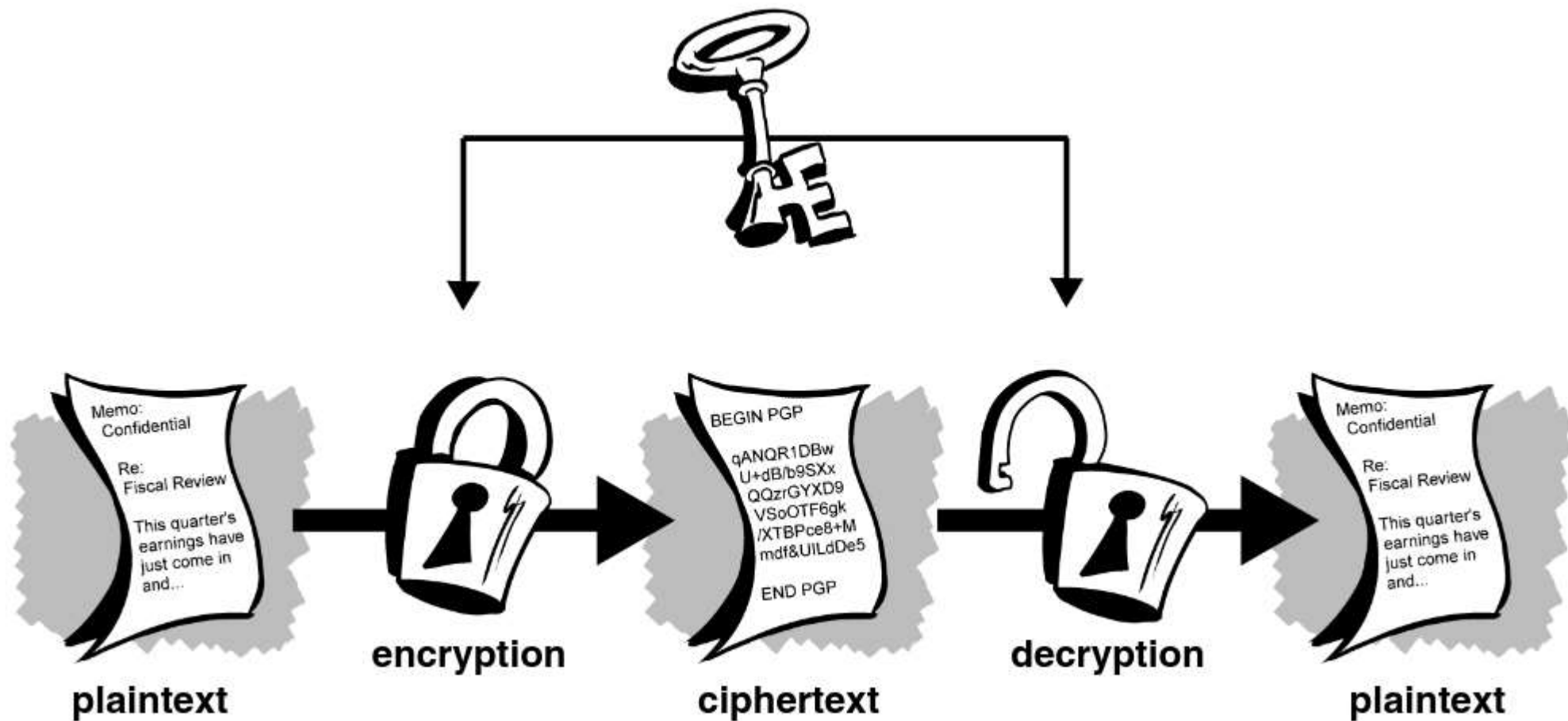
Encryption

- The words before making encryption is called plain text and after encryption is called cipher text
- To decrypt you must have the key to decrypt it as

Encryption



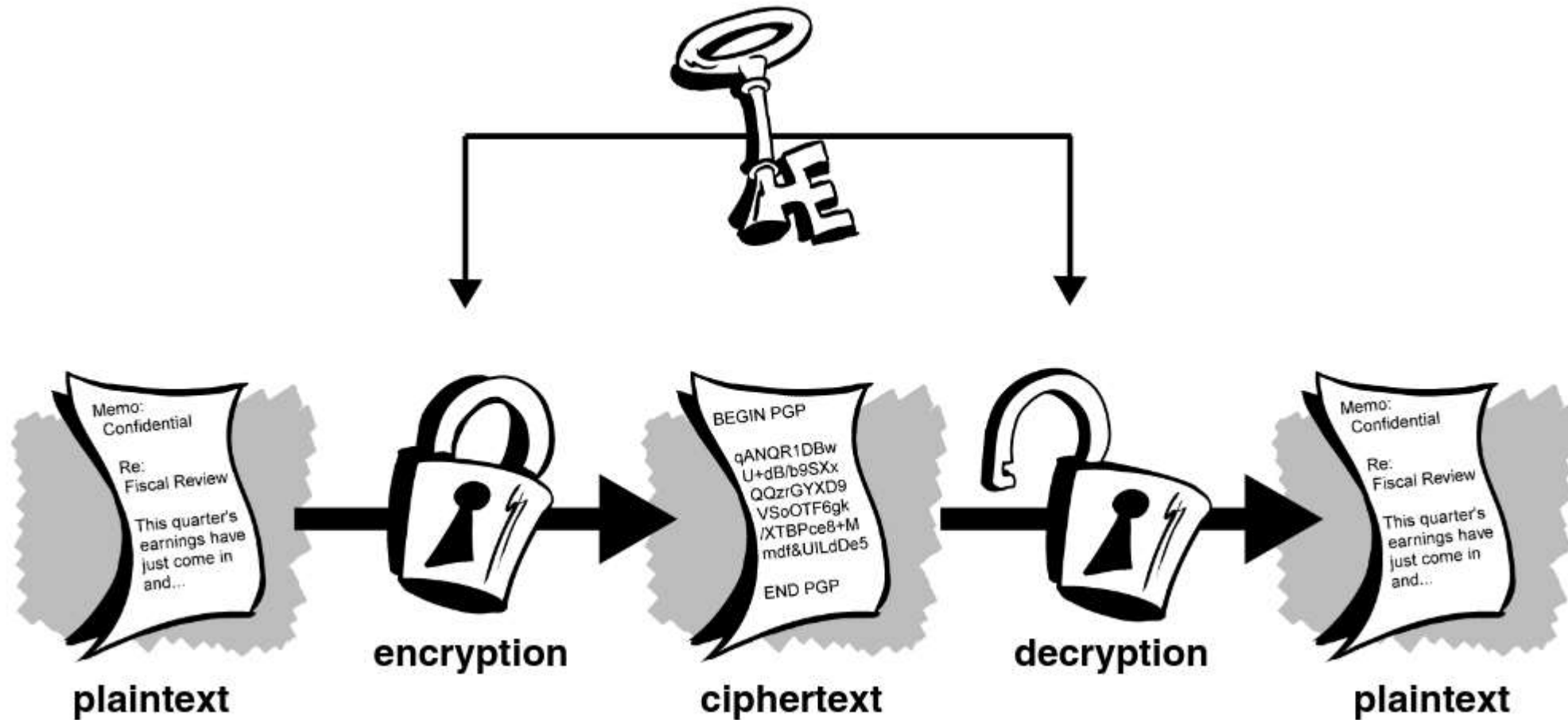
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Cryptography

- In cryptography you use the same key for encrypting and decrypting the data as in the previous image
- For example you have encrypted a message , your friend who would receive the message should have the same key to be able to decrypt the message and understand it
- This kind of encryption is called **Symmetric key**

Symmetric Key



Symmetric Key

The problem here is how can you send the key to your friend to decrypt the message that you encrypted

What if the hacker could steal a copy from that key while you are sending it

There are many problems here need to be solved

Symmetric Key Attack

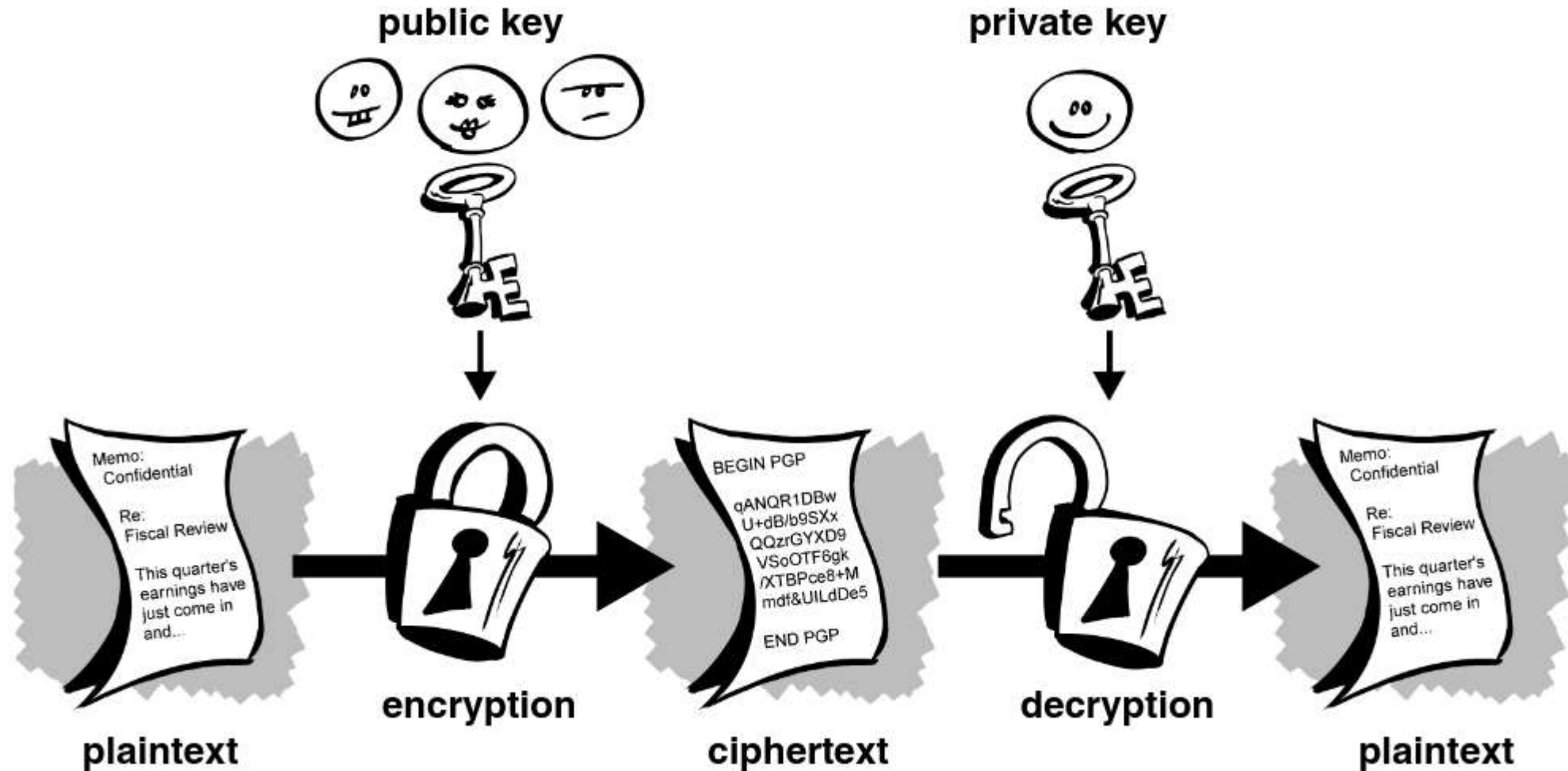
- Suppose that you are sending the key via message and attacker could see this message then he could decrypt all coming cipher messages easily
- To solve these problems invented **Asymmetric Keys**
- They are two keys that used to encrypt and decrypt
- And they are different

Asymmetric Keys

- Asymmetric Keys are two keys (Public Key and Private Key)
- Public Key is used to **encrypt** the data
- Private Key is Used to **Decrypt** the data

Asymmetric Keys

who has the corresponding private key can decrypt the information.



Asymmetric Keys

- Here the receiver should have Private key to be able to decrypt the message the the sender should have the public key to be able to encrypt the message
- Do you think what is more important the public key or the private key
- Another question : What do attackers search for public key or the private key ... think ...

Asymmetric Keys

- Absolutely Private key is the most important key as it is used to decrypt the message and if attacker could take it then he can decrypt and read all the messages easily
- Public key is not important for attacker anymore as all what this key do is to encrypt and attacker does not care about encryption he needs decryption

Asymmetric Keys

- Then How the problem of sending the message is solved ...
- Before you answer - you should know the two keys are generated together from one device

Asymmetric Keys

- The Solve is that : The receiving Person would generate the two keys and send the public key to his friend in public
- Ohh .. sending public key is not important for attacker

Asymmetric Keys



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```
(root@lulz-GlideFeed) - [~]  
# ssh-keygen -t rsa
```

Generating public/private rsa key pair.

Enter file in which to save the key (/sec/root/.ssh/id_rsa):

Enter passphrase (empty for no passphrase):

Enter same passphrase again:

Your identification has been saved in /sec/root/.ssh/id_rsa

Your public key has been saved in /sec/root/.ssh/id_rsa.pub

The key fingerprint is:

SHA256:UvgUb/MAZVqtRbv99qwTMzrCpg4/0gomRGyRQ4TNCe8 root@lulz-GlideFeed

The key's randomart image is:

```
+---[RSA 3072]-----+  
|.B+o      o.+o.|  
|.o*      . B  o.|  
|  =      . + =o.|  
|  +      + ..+ o|  
|  E      . S   o .|  
|  .      .      +.|  
|  . o .. .      +o|  
|    o ..oo + o .o.|  
|      .+++ . ..oo|  
+-----[SHA256]-----+
```

Hackers Techniques

- Question for you what do you think attackers use when creating Ransomware , Asymmetric Keys or Symmetric Keys
- Hackers use two keys to encrypt as they don't need Forensics engineers to get the key by analyzing the Ransomware and if engineers analyzed it they will find public key not the private key
- But they are searching for private key to decrypt the data , but it is owned only by attacker

Hash

- All what you need in hash is the algorithmics that you are gonna make the hash with
- Let's know more :
- There are many types of hash
- Md4,md5,sha1,sha256,sha512,bycrypt and more ...
- Every hash algothims has it's

Hash

- Difference between Hashing and encryption is that you need keys to encrypt the message
- But hash you don't need keys
- But how ,
- Every hash algorithms has format to know it with

Hash

- MD5 : 32 bytes (df2852a2b39ef0790c7acc806cdaca35)
- SHA1: 40 bits
(3dd29b9d75e470682695d3ca7ba2aa6c0536aced)
- SHA256: 64 bits
- SHA512: 128 bits
- Bcrypt: \$2a\$12\$rkEdwk/IIRmZPFUnVXCOP0YBaYrmQrCgwj3Ia
Sf5IXvX0vK8/43ju
-

Hash

- But wait , you didn't use the key to make hash then how would you decrypt it ...
- Actually the hash is not be decrypted it is fixed
- But if you need to get the plain text message you will need to crack this hash

Hash

- Cracking hash is to have very large table with many words and their hashes and then try every word as in picture in next slide
- Suppose that you have hash like this

3dd29b9d75e470682695d3ca7ba2aa6c0536aced

How would you crack it

Hash

3dd29b9d75e470682695d3ca7ba2aa6c0536aced

Before anything we know that hash is SHA1 as there are 40 bytes

Then ...

Hash



We need to crack 3dd29b9d75e470682695d3ca7ba2aa6c0536aced

Plain text	Hash (SHA1)
ali	b42a6d93d796915222f6ffb2ffdd6137d93c1cdb
12345@11	ff7bf06a610599df34afa8c92904d18f7595845d
123456789	f7c3bc1d808e04732adf679965ccc34ca7ae3441
01201331839	aa9ac46ce6321eabd4ad6f2120e8e201e1a98d5c
admin	fd0db110793105a640db0b3e4f73e6312b0ba9a5
kali123	4c0394bc580826e067765fbf618d2229010b130c
hossam	3dd29b9d75e470682695d3ca7ba2aa6c0536aced

Hash

- From the table found that the plain text is **hossam**

There are tools that help us to crack the hashes like John&ripper

And there are sites helps us to crack the hash as Crackstation.net

We will get back to hash and crack it in advanced level

Encoding

- Encoding is to change the Shape of words that can suites the place that is in for example:
- You are going to party you will wear T-shirt
- But if you are going to conference you will wear good suite with karvat
- But in all cases you are Ali and you can get back to your original shape as you were any time

Encoding

- Encoding has many shapes to suites the place that it is in as URL encoding , html encoding , base64 encoding etc...
- For us URL encoding , HTML Encoding , Base64 are the most important

Encoding

- Some examples for the formats of encoding:

- `< this is testing >`

url encoding: `%3Cthis%20is%20testing%3E`

HTML encoding: `<this is testing>`

Base64 Encoding : `PHRoaxMgaXMgdGVzdGluZyA+IA==`

Encoding

- Url encoding is always used in url links to make the browser know that it is a link
- Html Encoding is used in HTML language for sites
- Base64 Encoding is used while transferring the data with large size

Encoding

- You can decode any text by just searching for decode url or decode html or decode base64 etc...

Aslo you can use chatgpt for help any time

Encoding



- Any Questions